

Benchmarking General and Medical Small Language Models for Spanish Obstetric Question Answering: The Impact of Fine-Tuning

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Abstract—Small Language Models (SLMs) offer a computationally efficient alternative to large language models for clinical question answering; however, evidence on domain adaptation for Spanish obstetric applications remains limited. This study compares the general-purpose Gemma 4 model and the medically pretrained MedGemma 1.5 before and after grounded QLoRA adaptation using MaternaQA-es, while examining no retrieval, Hybrid retrieval, and independent HyDE as experimental factors. All 24 scenario-condition runs were evaluated across two 10-question scenarios, yielding 240 final condition-item outputs. Within this exploratory evaluation, Gemma 4 Base with Hybrid retrieval achieved the highest observed Answer Correctness and Semantic Similarity in both scenarios, whereas MedGemma 1.5 QLoRA without retrieval achieved the highest observed mean Answer Relevancy. HyDE generally improved retrieval precision relative to Hybrid, especially in the obstetric challenge scenario, but its effects on answer quality were heterogeneous and model-dependent. QLoRA improved all three no-retrieval response metrics for Gemma 4 in both scenarios and for MedGemma 1.5 in the obstetric challenge set, while the MaternaQA-es subset showed a correctness-relevance trade-off for MedGemma 1.5. Given the limited evaluation of 10 questions per scenario, these findings should be interpreted as preliminary descriptive patterns rather than definitive evidence of model or retrieval-strategy superiority. Some automatic metric values were missing, underscoring the need for broader evaluation and clinician review.

Index Terms—Small language models, QLoRA, retrieval-augmented generation, HyDE, obstetric question answering, Spanish clinical NLP, model benchmarking.

I. INTRODUCTION

The rapid adoption of Large Language Models (LLMs) has transformed the development of clinical decision-support systems and medical question-answering (QA) applications. Nevertheless, their computational requirements limit deployment in many healthcare environments, motivating increasing interest in Small Language Models (SLMs), which offer lower inference costs, reduced hardware requirements, and easier local deployment while maintaining competitive performance in domain-specific tasks [1], [2]. In medical applications,

supervised fine-tuning has emerged as a practical strategy for adapting compact language models to specialized clinical domains while preserving computational efficiency.

Recent studies have shown that domain-specific fine-tuning and retrieval can improve medical question answering under resource constraints [3]. Work on fine-tuned clinical embedding models likewise demonstrates the value of domain-specific retrieval, but remains outside Spanish obstetric QA [4]. Recent obstetric applications include simulated-case decision support, emergency-response evaluation, and neonatal growth prediction [5]–[7]. Reviews of maternal-health RAG emphasize the lack of standardized benchmarks and clinically validated evaluation protocols [8]. This gap is especially relevant in obstetric care, where inaccurate recommendations may affect maternal and fetal safety [9].

An important question therefore remains: does medical pretraining continue to provide measurable advantages after domain-specific fine-tuning? Medically pretrained SLMs should encode biomedical terminology, clinical concepts, and reasoning patterns. Yet, after both a general-purpose and a medical model receive identical specialized supervision, it remains unclear whether prior medical knowledge provides an additional benefit or whether the general-purpose model can achieve comparable or superior performance.

To investigate this question, this study uses MaternaQA-es, a Spanish obstetric QA resource containing 5,727 synthetic question-answer pairs generated from 63 Spanish-language clinical documents. The released document-level partitions contain 57 source PDFs: 52 in training, 2 in validation, and 3 in test. The dataset is publicly available through Hugging Face [10]. The benchmark compares Gemma 4 and MedGemma 1.5 before and after identical grounded QLoRA adaptation.

This work makes three contributions. First, it compares general-purpose and medically pretrained SLMs for Spanish obstetric QA. Second, it evaluates the effect of domain-specific

QLoRA adaptation under matched training conditions. Third, it provides a reproducible experimental design that incorporates retrieval-augmented inference and item-level evaluation while explicitly separating completed runs from planned factorial conditions.

II. METHODOLOGY

A. Study Design and Research Questions

The study uses a controlled, offline factorial benchmark. Model family has two levels: Google Gemma 4 E2B instruction-tuned (`google/gemma-4-E2B-it`) and Google MedGemma 1.5 4B instruction-tuned (`google/medgemma-1.5-4b-it`). Parameter adaptation has two levels: base weights without an adapter and grounded QLoRA. Retrieval strategy has three planned levels: the no-retrieval control (`no_rag`), canonical Hybrid retrieval, and canonical independent HyDE. Their Cartesian product defines 12 planned conditions.

The unit of analysis is an item-level paired `qa_id`. The primary QLoRA estimand is the paired difference between QLoRA and base outputs within a fixed model family and retrieval strategy. Retrieval contrasts compare strategies within a fixed family and adaptation state. The interaction estimand is the change in the QLoRA effect across retrieval strategies:

$$\Delta = (Y_{Q,H} - Y_{B,H}) - (Y_{Q,N} - Y_{B,N}). \quad (1)$$

Here Q denotes QLoRA, B denotes the base model, H denotes Hybrid retrieval, and N denotes the no-retrieval control. Analogous contrasts are defined for HyDE and for the two model families.

B. Dataset, Provenance, and Split Integrity

MaternaQA-es contains 5,727 Spanish obstetric question-answer pairs derived from 63 clinical documents. Its document-level partitions contain 52 training PDFs (5,093 items), 2 validation PDFs (306 items), and 3 held-out test PDFs. This study processed 10 questions from the test partition; test sources do not overlap training or validation sources, and `qa_id` is retained as the pairing key [10].

The primary grounded adaptation uses the `sft_grounded` representation, in which the source context associated with an item is concatenated with the question. The separate `sft_closed_book` representation contains only the question and is not a primary condition in the retained comparison. The term base indicates whether an adapter is loaded; it does not indicate that the input is closed-book when context is otherwise present.

The same 10 held-out test questions were processed across all model and retrieval conditions. The 2,268-chunk retrieval corpus includes the source material associated with these questions. Therefore, the RAG evaluation measures the system’s ability to retrieve and use relevant evidence when that evidence is available in the corpus, rather than strict generalization to completely unseen source documents. This inference-time access is distinct from leakage into model parameters, because

the corresponding test documents were not used during fine-tuning.

C. Protocol Evaluation Scenarios

Two protocol scenarios define the evaluation inputs. The first is a 10-item obstetric challenge set represented by `datasets/sample10.jsonl`; these records do not provide document-level reference chunk identifiers. The second consists of 10 processed questions from the held-out MaternaQA-es test partition, with QA metadata and source-reference information. MaternaQA-es is publicly available on Hugging Face [10].

The challenge set supports answer-quality evaluation but not gold document-level retrieval recall. The processed MaternaQA-es subset supports both response metrics and reference-chunk checks. Each scenario used 10 questions across all 12 conditions. MaternaQA-es preserves grounded source context with the question, whereas `sample10` contains only the question before retrieval-stage context is added. This input difference may affect the results.

Therefore, the two scenarios are treated as complementary evaluation settings rather than directly comparable samples from the same input distribution. Absolute metric differences between MaternaQA-es and the obstetric challenge set are not interpreted as effects of fine-tuning or retrieval, because scenario membership is confounded with differences in input construction and source-context availability. Model and retrieval comparisons are consequently emphasized within each scenario.

D. Model Families and Experimental Conditions

The benchmark compares Google Gemma 4 E2B instruction-tuned with Google MedGemma 1.5 4B instruction-tuned. Base conditions load the published model without an adapter. QLoRA conditions load `iue-edu/MaternaCare-ES-gemma4-qlora` or `iue-edu/MaternaCare-ES-medgemma-qlora` on the corresponding base model. The answer model, decoding settings, prompt structure, and test items are held fixed across retrieval strategies within a family and adaptation state.

Table I lists the complete prespecified design. The final result artifacts covered `no_rag`, Hybrid, and HyDE for all four model/adaptation combinations in both scenarios; therefore, all 24 scenario-condition runs were represented in the analysis.

E. Grounded QLoRA Fine-Tuning

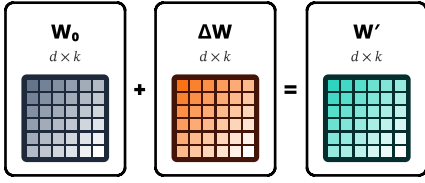
QLoRA [11] is the parameter-adaptation method in the benchmark: trainable low-rank adapters on quantized base weights, which keeps the base model as a separately identifiable initialization. Fig. 1 contrasts it with full fine-tuning, and Table II specifies the NF4 quantization and BF16 computation settings. RAFT is a related method for domain-specific RAG but is not an additional condition in this benchmark [12].

TABLE I
PRESPECIFIED FACTORIAL EXPERIMENT MATRIX

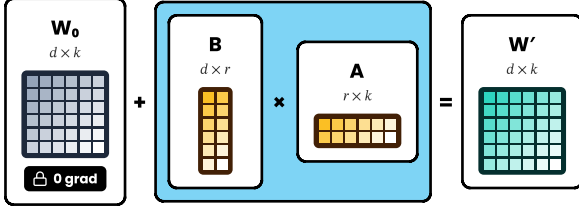
ID	Model family	Adaptation	Retrieval	Condition definition
C01	Gemma 4	Base	no_rag	Base weights; no retrieved context
C02	Gemma 4	Base	Hybrid	Base weights; canonical Hybrid retrieval
C03	Gemma 4	Base	HyDE	Base weights; canonical independent HyDE retrieval
C04	Gemma 4	QLoRA	no_rag	Grounded QLoRA adapter; no retrieved context
C05	Gemma 4	QLoRA	Hybrid	Grounded QLoRA adapter; canonical Hybrid retrieval
C06	Gemma 4	QLoRA	HyDE	Grounded QLoRA adapter; canonical independent HyDE retrieval
C07	MedGemma 1.5	Base	no_rag	Base weights; no retrieved context
C08	MedGemma 1.5	Base	Hybrid	Base weights; canonical Hybrid retrieval
C09	MedGemma 1.5	Base	HyDE	Base weights; canonical independent HyDE retrieval
C10	MedGemma 1.5	QLoRA	no_rag	Grounded QLoRA adapter; no retrieved context
C11	MedGemma 1.5	QLoRA	Hybrid	Grounded QLoRA adapter; canonical Hybrid retrieval
C12	MedGemma 1.5	QLoRA	HyDE	Grounded QLoRA adapter; canonical independent HyDE retrieval

Note. The complete design is 2 model families $\times$ 2 adaptation states $\times$ 3 retrieval strategies. All 12 conditions were observed in each of the two evaluation scenarios.

Full fine-tuning — every weight is trained



LoRA — freeze W_0 , train one low-rank branch



Trainable weights, same layer

full fine-tuning $\cdot d \cdot k$ — every weight

LoRA $\cdot r(d + k) \rightarrow$ orders of magnitude fewer (not to scale)

Fig. 1. Comparison of full fine-tuning and LoRA adaptation.

Both model families used the same MaternaQA-es training and validation partitions and the same grounded input representation. To maintain a controlled comparison between model initializations, the QLoRA hyperparameters and optimization settings were held constant across both models. Table II summarizes the training configuration used to produce the evaluated adapters.

The `sft_grounded` examples use the dataset’s conversational Contextofuente representation, in which source context is presented with the question. This differs from retrieval at inference: training context is pre-associated with each example, whereas RAG selects contexts from the 2,268-chunk corpus for each question. The adapter is loaded only in QLoRA conditions.

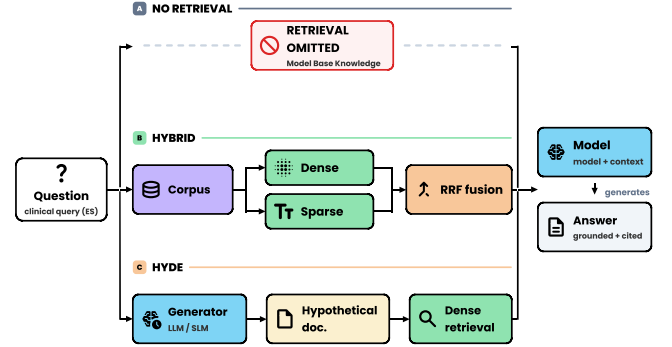


Fig. 2. RAG variants used in the benchmark: no retrieval, Hybrid retrieval with dense and sparse fusion, and independent HyDE retrieval.

F. Retrieval-Augmented Generation Protocol

RAG is the second core method [13]. The protocol treats retrieval strategy as a reproducible experimental factor. The no-retrieval control, Hybrid retrieval, and independent HyDE differ in how evidence is prepared before the answer model receives the question and context. Hypothetical text is used only for query expansion. Fig. ?? summarizes the three variants.

1) *Corpus Provenance and Chunk Construction*: The separate retrieval corpus contains 2,268 chunks from curated obstetric source documents and is used for dense, sparse, and fused retrieval. The `clinical_score` is a corpus-filtering signal, not clinical validation.

$$S(x) = \max\{0, L(x) + A(x) + B(W) - P(R)\}, \quad (2)$$

$$L(x) = \sum_{t \in T, t \subseteq \text{norm}(x)} [1 + \mathbf{1}(t \text{ contains a space})],$$

$$A(x) = \sum_{a \in A, a \subseteq \text{norm}(x)} 1,$$

$$B(W) = 21(W \geq 120) + 21(W \geq 350),$$

$$P(R) = 31(R \geq 0.35).$$

Here `norm(x)` applies Unicode normalization, removes combining marks, lowercases text, replaces non-ASCII letters and

TABLE II
QLoRA FINE-TUNING CONFIGURATION USED FOR BOTH MODEL FAMILIES

Parameter	Gemma 4 E2B QLoRA	MedGemma 1.5 4B QLoRA
Base model	google/gemma-4-E2B-it	google/medgemma-1.5-4b-it
Training representation	sft_grounded	sft_grounded
Train / validation examples	5,093 / 306	5,093 / 306
Quantization	4-bit NF4, double quantization	4-bit NF4, double quantization
LoRA rank (r)	16	16
LoRA alpha	16	16
LoRA dropout	0.05	0.05
Target modules	All linear (all-linear)	All linear (all-linear)
Learning rate	2×10^{-4}	2×10^{-4}
Batch size	1 per device	1 per device
Gradient accumulation steps	8	8
Effective batch size	8	8
Epochs / training steps	2 / 1,274	2 / 1,274
Optimizer	AdamW 8-bit	AdamW 8-bit
LR scheduler	Cosine	Cosine
Maximum sequence length	1,024	1,024
Packing	Disabled	Disabled
Random seed	3407	3407

Note. Both model families used the same training and validation partitions, grounded input representation, and QLoRA optimization settings. The exact mixed-precision mode of the retained training runs could not be reliably recovered and is therefore not reported.

punctuation with spaces, and collapses whitespace. T is the clinical-term set, A the action-term set, W the Unicode word count, and R the ratio of reference-matching nonempty lines to all nonempty lines.

2) *Dense and Sparse Retrieval:* Dense retrieval uses BAAI / bge-m3 [14]. SentenceTransformer produces normalized dense vectors on CPU with `batch_size=32`. The persistent local vector table is LanceDB. Because no approximate vector index is created, search is exhaustive exact cosine search. The default dense retrieval depth is $k = 5$.

Sparse retrieval is Okapi BM25 (`rank_bm25 / BM25Okapi`) over Unicode lowercase word tokens. Retrieval outputs are recorded before answer generation so retrieval quality and latency can be analyzed separately from response generation.

3) *Canonical Hybrid Retrieval:* Hybrid retrieval fuses dense and BM25 rankings by reciprocal rank fusion with `rank_constant=60` [15]. For a requested final depth k , the candidate depth is:

$$k_c = \min(|C|, \max(4k, 50)). \quad (3)$$

Ties are broken by stable chunk ID, and the final top- k chunks are supplied to the answer model.

4) *Canonical Independent HyDE Retrieval:* HyDE [16] uses a dedicated hypothetical-document generator, separate from the answer model, to produce a Spanish clinical-document passage from the question. The retained runs used the OpenAI provider with `gpt-5-mini-2025-08-07`. Hypothetical documents were generated once per question and cached for reuse across answer-model conditions; dense retrieval then embeds that text with BGE-M3 and searches the corpus by exact cosine similarity. Only the retrieved corpus chunks, not the hypothetical text, are supplied as evidence to the answer model [17]. The cache is separated by dataset mode, provider, and generator model, and row reuse checks `qa_id` and a question hash. The request-side output limit could not be recovered from the retained run records.

5) *No-Retrieval Control:* The no-retrieval control (`no_rag`) returns no retrieved contexts and omits the retrieved-context block from the answer prompt. Context-dependent retrieval metrics are therefore not defined and are not assigned a value of zero.

G. Inference and Prompt Controls

All answer prompts are written in Spanish. RAG prompts contain a clearly delimited context block with positional labels and the question. Provenance is retained in benchmark records rather than asserted as content in the answer prompt. The `no_rag` condition receives the same question formatting and chat template but an empty context block.

Inference uses the model’s chat template and greedy decoding by default, with a maximum of 512 newly generated tokens. Sampling is disabled unless a run explicitly records a different configuration. QLoRA conditions load the declared adapter on the corresponding base model; base conditions do not. Retrieval generation and answer generation remain separate stages.

H. Evaluation Framework and Metrics

RAGAS 0.4.3 [18] is used as an external automatic evaluation layer. It is neither the retriever nor a clinical expert. Response-quality, retrieval-quality, and operational measurement domains are kept separate before paired contrasts and interaction analysis. Figure ?? illustrates the RAGAS evaluation workflow, showing how each part contributes to scoring the responses.

1) *Retained Fine-Tuning-Output Evaluator:* The retained evaluator uses RAGAS 0.4.3 with `gpt-5.4-mini` as evaluator, `text-embedding-3-small` for embedding-based evaluation, and a maximum completion length of 2,048 tokens. The retained metrics are Faithfulness, Answer Relevancy, Answer Correctness, and Semantic Similarity.

2) *RAG Benchmark Evaluator:* The RAG evaluator uses six metrics: Context Precision, Context Recall, Faithfulness, Answer Relevancy, Answer Correctness, and Semantic Similarity.

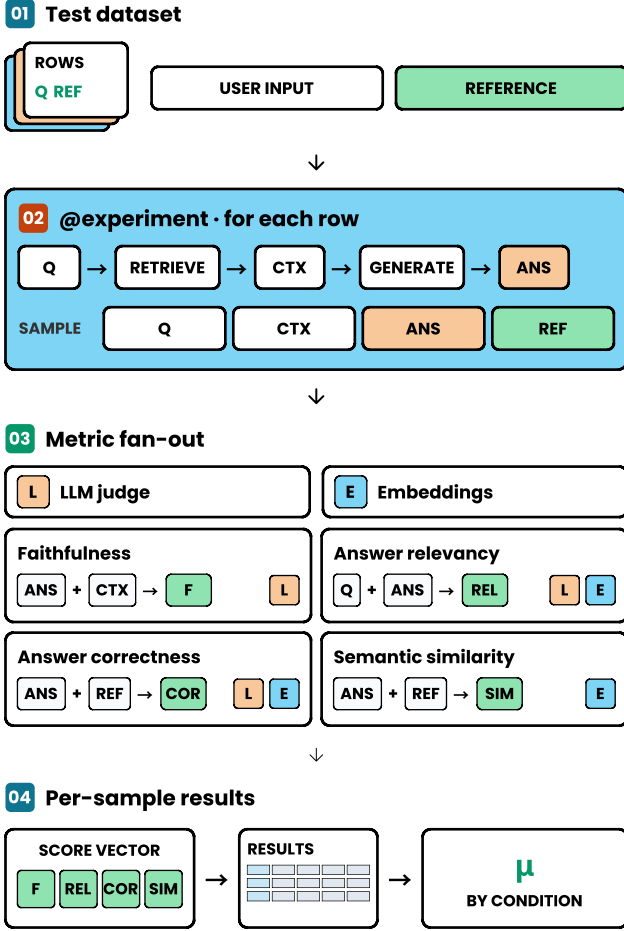


Fig. 3. RAGAS response-evaluation workflow from per-question inputs and evidence to per-condition results.

Their definitions follow RAGAS and are not restated here; the scoring pipeline is summarized in Fig. ??.

Operational variables include input, output, and total token counts; retrieval, generation, and end-to-end latency; and output tokens per second. Wall-clock output rate is descriptive and not treated as hardware-independent throughput. For `no_rag`, Context Precision, Context Recall, and Faithfulness are missing by design rather than zero.

$$\text{TPS} = \frac{N_{\text{out}}}{T_{\text{gen}}}. \quad (4)$$

I. Analysis Plan

The primary analysis is descriptive and item-level. Means are computed from the valid automatic scores available within each balanced 10-question scenario; missing evaluator values are not imputed. Given the limited number of evaluated questions per scenario, all comparisons are treated as exploratory rather than confirmatory. QLoRA is compared with base within each retrieval strategy and family; retrieval strategies are contrasted within adaptation and family; and model families are compared under matched conditions.

Duplicate or superseded records were resolved at `qa_id` level by preferring more complete rows, fewer metric errors, and then the latest row position. The summary used the same 10 `qa_id` values shared by all 12 conditions in each scenario.

J. Reproducibility and Experimental Scope

Reproducibility requires recording the MaternaQA-es revision, retrieval-corpus hash, complete configuration, prompts, model and adapter identifiers, HyDE provider settings, cache identity, evaluator version, and decoding settings. The pinned BGE-M3 revision and deterministic RRF tie-breaking are part of this record. Reporting follows the transparency goals of TRIPOD-LLM and Data Statements for NLP [19], [20].

QLoRA fine-tuning and local neural-model computations were performed on an NVIDIA GeForce RTX 5070 Ti Laptop GPU with 12 GB of dedicated VRAM. The comparison comprised 20 questions—10 from MaternaQA-es and 10 from `sample10`—evaluated across all 12 conditions, and all principal comparisons are made within each scenario.

III. RESULTS

A. Available Runs and Analysis Set

The consolidated workbook contained 262 records, including duplicate, retrieved, or superseded rows, and represented two scenarios, four model conditions, and all three inference strategies. `no_rag`, Hybrid, and HyDE were available for every model/adaptation combination in both scenarios, completing the 12-condition matrix for each scenario.

After resolving duplicate or superseded rows and restricting both scenarios to identifiers shared by all 12 conditions, the balanced analysis contained 240 final condition-item outputs: 120 for the MaternaQA-es subset and 120 for the obstetric challenge set. Every condition contributed responses for all 10 questions; however, 20 condition-item records contained at least one unavailable automatic metric, so means were calculated from valid scores without imputation.

B. MaternaQA-es Subset

Gemma 4 Base with Hybrid retrieval achieved the highest observed mean Answer Correctness (0.668) and Semantic Similarity (0.813), while MedGemma 1.5 QLoRA without retrieval obtained the highest observed mean Answer Relevancy (0.811). Under no retrieval, QLoRA increased Gemma 4 Answer Correctness by 0.129, Answer Relevancy by 0.047, and Semantic Similarity by 0.053. For MedGemma 1.5, QLoRA reduced Answer Correctness by 0.057 but increased Answer Relevancy by 0.212 and Semantic Similarity by 0.109.

Relative to `no_rag`, Hybrid and HyDE produced their largest response-quality gains for Gemma 4 Base: Answer Correctness increased by 0.348 with Hybrid and 0.336 with HyDE. HyDE also increased Answer Correctness for MedGemma 1.5 Base (+0.107) and QLoRA (+0.077), but reduced it for Gemma 4 QLoRA (−0.062). Directly against Hybrid, HyDE improved Answer Correctness only for MedGemma 1.5 QLoRA (+0.042) and improved Answer Relevancy most clearly for MedGemma 1.5 Base (+0.135), illustrating heterogeneous retrieval effects.

TABLE III
MEAN AUTOMATIC METRICS FOR THE 10 SHARED QUESTIONS IN EACH SCENARIO

Model	State	Strategy	Correct.	Relev.	Sem. sim.	Faith.	C. prec.	C. recall
<i>Panel A. MaternaQA-es test subset</i>								
Gemma 4	Base	No RAG	0.320	0.689	0.684	—	—	—
Gemma 4	Base	Hybrid	0.668	0.786	0.813	0.917	0.945	0.900
Gemma 4	Base	HyDE	0.656	0.710	0.802	1.000	0.964	0.850
Gemma 4	QLoRA	No RAG	0.449	0.736	0.737	—	—	—
Gemma 4	QLoRA	Hybrid	0.451	0.728	0.791	0.768	1.000	0.900
Gemma 4	QLoRA	HyDE	0.387	0.643	0.699	0.904	0.933	0.850
MedGemma 1.5	Base	No RAG	0.367	0.599	0.625	—	—	—
MedGemma 1.5	Base	Hybrid	0.499	0.562	0.607	0.790	0.945	0.850
MedGemma 1.5	Base	HyDE	0.474	0.697	0.631	0.869	0.958	0.850
MedGemma 1.5	QLoRA	No RAG	0.311	0.811	0.733	—	—	—
MedGemma 1.5	QLoRA	Hybrid	0.346	0.706	0.749	0.841	0.900	0.812
MedGemma 1.5	QLoRA	HyDE	0.388	0.649	0.754	0.761	0.958	0.900
<i>Panel B. Obstetric challenge set</i>								
Gemma 4	Base	No RAG	0.206	0.708	0.620	—	—	—
Gemma 4	Base	Hybrid	0.481	0.692	0.736	0.986	0.760	0.550
Gemma 4	Base	HyDE	0.423	0.760	0.722	0.960	0.917	0.800
Gemma 4	QLoRA	No RAG	0.268	0.824	0.648	—	—	—
Gemma 4	QLoRA	Hybrid	0.370	0.647	0.672	0.898	0.796	0.500
Gemma 4	QLoRA	HyDE	0.368	0.676	0.695	0.894	0.898	0.600
MedGemma 1.5	Base	No RAG	0.216	0.833	0.620	—	—	—
MedGemma 1.5	Base	Hybrid	0.240	0.388	0.509	0.930	0.811	0.500
MedGemma 1.5	Base	HyDE	0.368	0.607	0.535	0.885	0.908	0.700
MedGemma 1.5	QLoRA	No RAG	0.274	0.881	0.652	—	—	—
MedGemma 1.5	QLoRA	Hybrid	0.373	0.741	0.685	0.590	0.841	0.714
MedGemma 1.5	QLoRA	HyDE	0.256	0.696	0.660	0.764	0.919	0.700

Note. Correct. = answer correctness; Relev. = answer relevancy; Sem. sim. = semantic similarity; Faith. = faithfulness; C. prec. = context precision; C. recall = context recall. Dashes mark non-applicability for no_rag. All conditions yielded 10 responses; valid n was 5–10 for response metrics and 6–10 for retrieval metrics because some evaluator outputs were unavailable. No missing value was imputed; bold values are panel maxima.

C. Obstetric Challenge Set

Within the evaluated obstetric challenge subset, Gemma 4 Base with Hybrid retrieval achieved the highest observed mean Answer Correctness (0.481) and Semantic Similarity (0.736), whereas MedGemma 1.5 QLoRA without retrieval obtained the highest observed mean Answer Relevancy (0.881). Under no retrieval, QLoRA improved all three response metrics for both model families.

Relative to no_rag, Hybrid increased Answer Correctness for all four model conditions but reduced Answer Relevancy in every condition. Semantic Similarity increased for both Gemma 4 conditions and MedGemma 1.5 QLoRA, but decreased for MedGemma 1.5 Base.

Relative to no_rag, HyDE increased Answer Correctness for Gemma 4 Base (+0.218), Gemma 4 QLoRA (+0.100), and MedGemma 1.5 Base (+0.152), but slightly reduced it for MedGemma 1.5 QLoRA (−0.018). Answer Relevancy increased only for Gemma 4 Base, while Semantic Similarity increased for both Gemma 4 conditions and MedGemma 1.5 QLoRA but decreased for MedGemma 1.5 Base.

Compared directly with Hybrid retrieval, HyDE improved Answer Relevancy for both Gemma 4 conditions and MedGemma 1.5 Base, and it improved Answer Correctness most clearly for MedGemma 1.5 Base (+0.127). It did not dominate Hybrid for the other model conditions, confirming that hypothetical-document expansion changed retrieval quality without guaranteeing better final answers.

D. Retrieval-Grounding Metrics

In the MaternaQA-es subset, Hybrid Context Precision ranged from 0.900 to 1.000, Context Recall from 0.813 to 0.900, and Faithfulness from 0.768 to 0.917. HyDE Context Precision ranged from 0.933 to 0.964, Context Recall from 0.850 to 0.900, and Faithfulness from 0.761 to 1.000. HyDE exceeded the corresponding Hybrid Context Precision in three of four model states, but improved Context Recall in only one state.

In the obstetric challenge set, Hybrid Context Precision ranged from 0.760 to 0.841, Context Recall from 0.500 to 0.714, and Faithfulness from 0.590 to 0.986. HyDE Context Precision ranged from 0.898 to 0.919, Context Recall from 0.600 to 0.800, and Faithfulness from 0.764 to 0.960. HyDE achieved higher Context Precision than Hybrid for all four model states and higher Context Recall for three of four states, although these retrieval gains were not accompanied by uniform response-quality gains.

Across both scenarios, the 20 unique uncached HyDE generations averaged 18.5 seconds, 67.1 input tokens, and 1,437.3 API output tokens. The same hypothetical documents were then reused from cache across the remaining answer-model conditions, reducing model-dependent variation in the query-expansion stage.

Faithfulness, Context Precision, and Context Recall were not calculated for no_rag because those conditions did not supply retrieved evidence. Their absence represents non-applicability rather than zero performance.

E. Run and Metric Coverage

All 24 scenario-condition runs generated responses for their 10 shared questions; therefore, there were no condition-level execution failures in the retained matrix. The workbook nevertheless contained unavailable automatic scores in 20 of the 240 selected condition-item records (26 missing applicable metric cells in MaternaQA-es and 31 in the obstetric challenge set). These values were left missing rather than imputed, and Table III reports means over the valid scores available for each metric. Context-dependent metrics remained not applicable to no_rag by design.

IV. DISCUSSION

Across the evaluated 20-question benchmark, the observed effect of QLoRA varied according to both model initialization and inference strategy. In no-retrieval conditions, QLoRA improved Answer Correctness, Answer Relevancy, and Semantic Similarity for Gemma 4 in both scenarios. MedGemma 1.5 showed the same pattern in the obstetric challenge set, but the MaternaQA-es subset exhibited a trade-off: correctness decreased while relevance and semantic similarity increased.

Within the evaluated questions, medical pretraining did not show a uniform advantage after domain-specific adaptation across the tested metrics and configurations. MedGemma 1.5 QLoRA achieved the strongest Answer Relevancy without retrieval in both scenarios, suggesting that medical initialization may support focused clinical responses. However, Gemma 4 Base with Hybrid retrieval achieved the best Answer Correctness and Semantic Similarity in both scenarios. These findings support evaluating multiple dimensions rather than treating a single automatic score as a complete measure of quality.

Hybrid retrieval generally improved Answer Correctness but often reduced Answer Relevancy. One plausible interpretation is that retrieved evidence increased overlap with reference content while encouraging longer or less directly focused responses. Retrieval-grounding scores were generally high for precision but more variable for recall, indicating that relevant evidence was often surfaced without consistently covering all reference-supporting information.

HyDE produced consistently high retrieval precision in both scenarios, but its effects on final answers were heterogeneous. In MaternaQA-es, it improved Context Precision relative to Hybrid for three model states yet improved Answer Correctness over Hybrid only for MedGemma 1.5 QLoRA. In the obstetric challenge set, it improved Context Precision for all four states and recovered substantially more Answer Correctness than Hybrid for MedGemma 1.5 Base. Better query expansion and evidence ranking therefore appear to be enabling factors rather than sufficient conditions for better answers.

V. CONCLUSIONS

In this exploratory 20-question benchmark, grounded QLoRA improved the no-retrieval response metrics of Gemma 4 across both scenarios, while its effect on MedGemma 1.5 was more context-dependent. Medical pretraining did not show a consistent performance advantage across the tested conditions

after identical obstetric adaptation. Hybrid retrieval improved Answer Correctness in all four obstetric-challenge comparisons and in three of four MaternaQA-es comparisons, but it could reduce Answer Relevancy, demonstrating a trade-off between reference agreement and directness.

HyDE completed all planned conditions in both processed scenarios and generally produced high Context Precision, but it did not consistently improve Answer Correctness, Answer Relevancy, or Semantic Similarity over Hybrid retrieval. The evidence therefore supports a nuanced conclusion: observed benefits varied by model family, metric, and scenario. Because these are descriptive patterns from 10 questions per scenario, broader question coverage, complete evaluation, and clinician assessment remain necessary before drawing deployment-oriented conclusions.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES

The authors used ChatGPT [21] only to support language editing, code and reference review, figure preparation, and document formatting. It was not used to generate, modify, or evaluate experimental data or to make clinical judgments. All AI-assisted outputs were reviewed by the authors, who take responsibility for the final content.

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